

Unit 11, Lesson 8: Triangle Midsegment Theorem



Benchmarks

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can discover the triangle midsegment theorem using AA similarity.
- ☐ I can use the triangle midsegment theorem to prove relationships and to solve problems.



11.8.1 Warm-Up: "Many Small Cubes"

1. Japanese architect Sou Fujimoto exhibited a sculpture of stacked and suspended aluminum cubes as a part of the art fair in the Parisian Jardins des Tuileries' gardens.

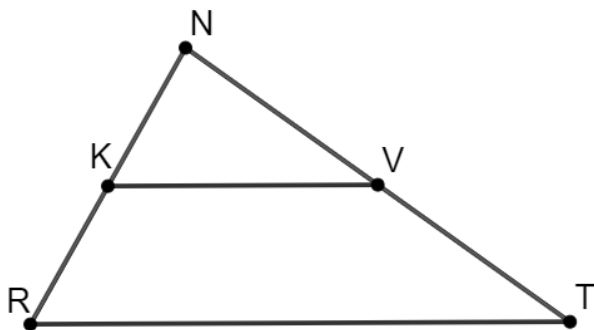


- a. What do you notice about the structure?
- b. What do you wonder?



11.8.2 Exploration: Triangle Midsegment Theorem

$\triangle NKV$ and $\triangle NRT$ are shown, where $\overline{KV} \parallel \overline{RT}$.



1. Discuss with your partner what can be concluded from the given information. Justify your conclusions.
2. Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

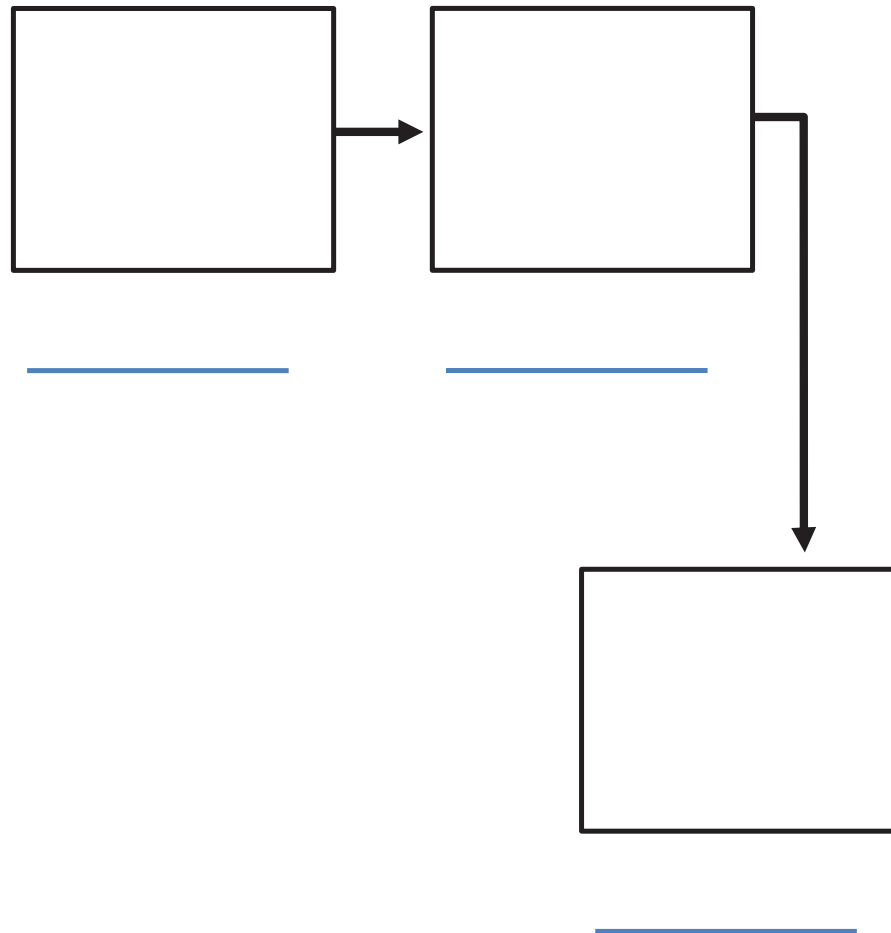
Conclusion	My Summary of Their Reason	Initials



3. Given: $\triangle NRT$ with $\overline{KV} \parallel \overline{RT}$

Prove: $\triangle NKV \sim \triangle NRT$

Complete the flowchart proof.



4. What other information about triangles NKV and NRT can be determined from knowing that $\triangle NKV \sim \triangle NRT$?

5. Complete the table using the information $KV = 5$, $NR = 6$, $NK = 3$, and $NT = 8.8$.

$RT =$	$NV =$
$KR =$	$VT =$

6. What do you notice about the location of $\overline{KV}$?
7. What do you notice about the lengths of $\overline{KV}$ and $\overline{RT}$?



Triangle Midsegment Theorem

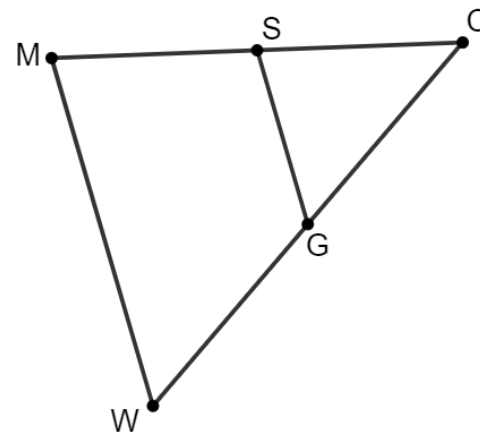
The midsegment of a triangle joins the midpoints of two sides of a triangle such that its length is half the length of the third side of the triangle, and it is parallel to the third side of the triangle.



11.8.3 Guided Practice: Triangle Midsegment Theorem

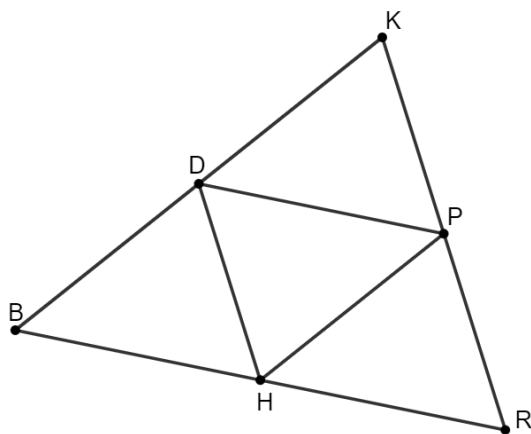
1. $\triangle MCW$ is shown, where $\overline{SG}$ is the midsegment.

$$\begin{aligned} MS &= \left(\frac{1}{2}x + 4\right), \\ GC &= (11z - 2), \\ SG &= (2y + 5), \\ MW &= (8y - 2), \\ SC &= \left(\frac{3}{4}x - \frac{1}{4}\right), \\ WG &= (9z + 1). \end{aligned}$$



- Find the value of x .
- Find the value of y .
- Find the value of z .
- What is the perimeter of $\triangle MCW$?

2. $\triangle BKR$ is shown, where D , P , and H are the midpoints and the perimeter of $\triangle DPH$ is 23.5. $DP = 17x$, $PH = (6x + 5)$, and $HD = 20x - 3$.

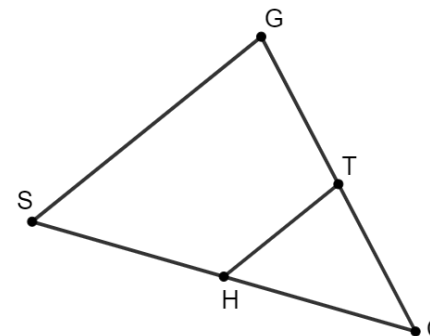


a. What is the value of x ?

b. Complete the table.

$BK =$	$DH =$
$BR =$	$KP =$
$DP =$	$KR =$

3. $\triangle SGC$ and $\triangle HTC$ are shown.



Given: $SH = HC$ and $GT = TC$

Prove: $\overline{SG} \parallel \overline{HT}$

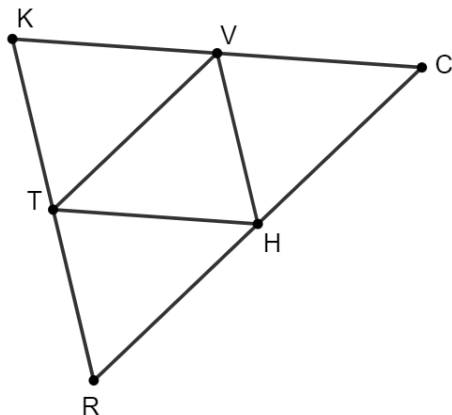
Since $SH = HC$ and $GT = TC$, point H is the _____ of $\overline{SC}$ and point T is the _____ of $\overline{GC}$ by _____. Therefore, $\overline{HT}$ is the _____ of $\triangle SGC$ by definition and $\overline{SG} \parallel \overline{HT}$ by the _____.



11.8.4 Your Turn!

$\triangle KCR$ is shown, where T , V , and H are midpoints.

$KC = (5x)$, $VH = (\frac{1}{3}y + 11)$, $VT = (6w - 5)$, $KR = (2y - 2)$, $RC = (8w + 8)$, and $TH = (3x - 4)$.



- Find the value of each variable.

$x =$	$y =$	$w =$
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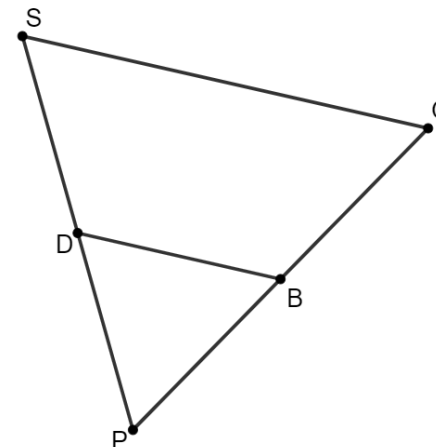
- Complete the table.

$TH =$	$KC =$
$VH =$	$KR =$
$VT =$	$RC =$



11.8.5 Wrap-Up: Error Analysis

- $\triangle SGP$ is shown, where $\overline{DB}$ is a midsegment. $SG = 18x + 22$ and $DB = 11x + 7$.



Priscilla was finding the value of x . Her work is shown.

Priscilla's Work

$$\begin{aligned} 18x + 22 &= 11x + 7 \\ 7x &= -15 \\ x &= -\frac{15}{7} \end{aligned}$$

- Write a note to Priscilla explaining her mistake.
- Find the value of x .